

Towards a Design Space for Emotion Recognition

Vivian Genaro Motti

vmotti@gmu.edu

George Mason University

Fairfax, Virginia, USA

ABSTRACT

Recognizing emotions through pervasive technologies has been the objective of several research studies. Still, the inherent complexities of emotion recognition result in applications that support the process rather than fully address it. Existing methods for data collection combine multiple sensors and devices. The analysis and classification of users' and environmental data face challenges to reach high accuracy, responsiveness, and reliability. Because emotion recognition is user-dependent, it still remains as an open challenge to be resolved. This paper provides a unified view about emotion recognition, presenting a multidimensional design space that includes multiple approaches. We discuss the practicalities involved with the implementation of each approach and the deployment of emotion recognition systems. We conclude with a discussion about their advantages, limitations, and open questions.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile devices**.

KEYWORDS

affective computing; emotion recognition; design space; comparative analysis

ACM Reference Format:

Vivian Genaro Motti. 2021. Towards a Design Space for Emotion Recognition. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*, September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3460418.3479337>

1 INTRODUCTION

Enabling emotion recognition through technology has proven to be important for several applications across domains, benefiting diagnostics, monitoring, assistance, and therapeutic interventions. In education, and online learning platforms, emotion recognition facilitates the assessment of students' engagement, enabling applications that adapt their contents accordingly to ensure better learning outcomes. In interactive art installations, the recognition of affective states can use results from the analysis of the user experience

to tailor exhibitions according to the users' emotional responses [11]. In automotive industry, emotion recognition coupled with behavioral interventions can help drivers to reduce their stress levels as well as the risk of car accidents [24, 25, 32]. Emotion recognition has a large range of applicability, being also studied for healthcare delivery [3, 19], mental health [26], assistive technologies [33], and games [15].

Pervasive solutions for emotion recognition with wearable devices have the advantage of being unobtrusive and more reliable than external observations, thanks to objective measurements with sensors that are not invasive. Still, to be reliable emotion recognition faces several practical challenges that cut across sensing and interventions [18]. Additionally, no universal model that is suitable for diverse users' profiles exist [5]. However, recognition efforts must be tailored to meet individual's needs, accounting for users' contexts and cultural aspects. In practice, deciding what are the most appropriate and accurate sensors requires in-depth understanding of the problem space and users' context [21]. The calibration of the classification models must take into account the baseline measurements of each user, as well as specific cultural differences. Furthermore, there is a wide variety of sensors that can be employed to collect physiological signals that serve as indicators for emotional states [20]. The placement of such sensors and the choice of form factors that house them lead to trade-offs regarding comfort levels, accuracy, and energy-efficiency [22]. Also, the analysis of the data collected requires a thorough consideration of often conflicting quality criteria, such as accuracy and responsiveness. To optimize the analysis process, sampling rates must be reduced, saving costs related to computational power, memory for storage, and data transmission. Considerations about privacy, ethics, and fairness also gained increased attention in recent studies [1]—posing important challenges to the deployment and widespread adoption of emotion recognition systems that collect personal data continuously and inconspicuously in the wild.

Several approaches have been studied to recognize emotions. In general, as Figure 1 shows, the process of emotion recognition begins with a data collection step, followed by pre-processing and feature extraction, and completed with data analysis for the classification of emotional states. In case interventions are involved, the emotion classification feeds a recommendation system, and the intervention is delivered for end users seeking to help them with emotion regulation. In terms of data analysis specifically, statistical techniques as well as artificial intelligence and machine learning models have been widely investigated. Their application and success rely heavily on the amount of data available and their characteristics (sparsity, volume, reliability). Techniques such as regression analysis, Support Vector Machines (SVM) Predictive

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA

© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-8461-2/21/09...\$15.00
<https://doi.org/10.1145/3460418.3479337>

and Descriptive Models, Decision Trees, Clustering, Nearest Neighbors (KNN), Naive Bayes, Random Forest, and Neuronal Networks exemplify how data analysis has been approached.

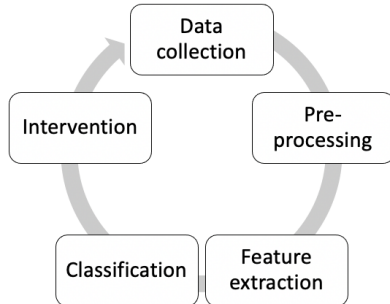


Figure 1: The data analysis process for emotion recognition requires pre-processing after data collection, which is followed by extraction of features and classification. When interventions are pre-defined, the model feeds a recommendation system and the interventions are delivered to end users. If feedback from end users is sought, the process follows a cyclic approach until the aimed objectives are achieved.

The recognition of affective states using wearable sensors has become increasingly important to address mental health crises. More specifically, the development and deployment of affective computing applications addresses a growing need for interventions, facilitating the delivery of mental health services. These applications are beneficial for all stakeholders involved, reducing the costs associated with healthcare delivery, increasing access to services, and reducing potential stigmas as well. Mobile sensing through wearables, smartphones, and pervasive computing enables applications to sense, monitor, and recognize users' affective states in a continuous way, besides also delivering personalized interventions on demand and generating data sets for aggregated analysis in large scale.

This paper presents an overview of the approaches employed for emotion recognition. First, we reviewed the literature to contextualize the research and practice on data analysis for emotion recognition. Then, we describe data sources, signals and sensors used in affect recognition. We discuss the statistical analysis approaches, machine learning techniques and models used, as well as the tools and frameworks employed to classify data from users and their environments aiming to recognize emotional states. Lastly, we conclude with a discussion about ethics, fairness, bias as well as the trade-offs and evaluation criteria involved in each approach.

2 DESIGN SPACE

A growing body of research has been dedicated to study emotion recognition. In this section we present the results of the analysis of the literature, categorizing previous contributions in: (1) theoretical models, (2) platforms, and (3) approaches (sensors and data). This section aims to provide a unified design space of emotion recognition (Figure 2).

Theory	Description	Ref
Ekman's Atlas	is a tool to help people better understand what emotions are, how they are triggered and what their effects are	[9]
PAD Model	is an emotional state model developed by Albert Mehrabian and James A. Russell (1974) to describe emotional states; PAD uses three numerical dimensions: Pleasure, Arousal and Dominance to represent all emotions	[17]
Geneva Emotion Wheel (GEW)	is a theoretically derived and empirically tested instrument to measure emotional reactions to objects, events, and situations	[27]

Table 1: Theoretical models for emotion recognition

2.1 Theoretical Models

In order for computational solutions to effectively recognize emotions, an informed and integrated examination of related theories is necessary. In addition, multiple areas must be considered in an implementation process if real-world systems are to be achieved [6]. As listed in Table 1, theoretical frameworks related to emotion recognition, include Ekman's atlas of emotions [9] as well as Scherer's emotion wheel [27]. Mehrabian's Pleasure-Arousal-Dominance (PAD) model is an affective model of user experience [17]. It measures emotional tendencies and response along three dimensions: pleasure-displeasure, corresponding to cognitive evaluative judgements; arousal-non-arousal to refer to levels of alertness and physical activity; and dominance-submissiveness to represent the feeling of control and influence over others and surroundings. These dimensions describe emotions broadly, distinguishing basic emotion categories, as well as emotions that more common two-dimensional models conflate (e.g., anger vs. fear) [11].

2.2 Platforms

Among the ecosystem of devices studied in emotion recognition, we highlight tablets [10], smartphones, wearables, robots [16], and instrumented environments, including personal assistants and smart cars [32]. The widespread availability of smartphones enabled emotion recognition studies that are longitudinal, inexpensive, and deployed in real-world scenarios. Mobile sensors enable self-assessment from users [4], as well as passive sensing from their context [23]. Stress detection and prediction benefit from mobile approaches [14].

Among wearable devices, form factors include wrist-worn wearables, with commercial devices such as the E4, Feel, Upmood and Amazon Halo, as well as customized prototypes and applications employed in large scale research projects (e.g., StudentLife [12, 29], EXP [31]). Regarding chest bands and other body-area-network devices, we highlight Polar for heart rate monitoring, head-mounted devices such as glasses for affect recognition through detection of facial expressions, and VR helmets to deliver interventions.

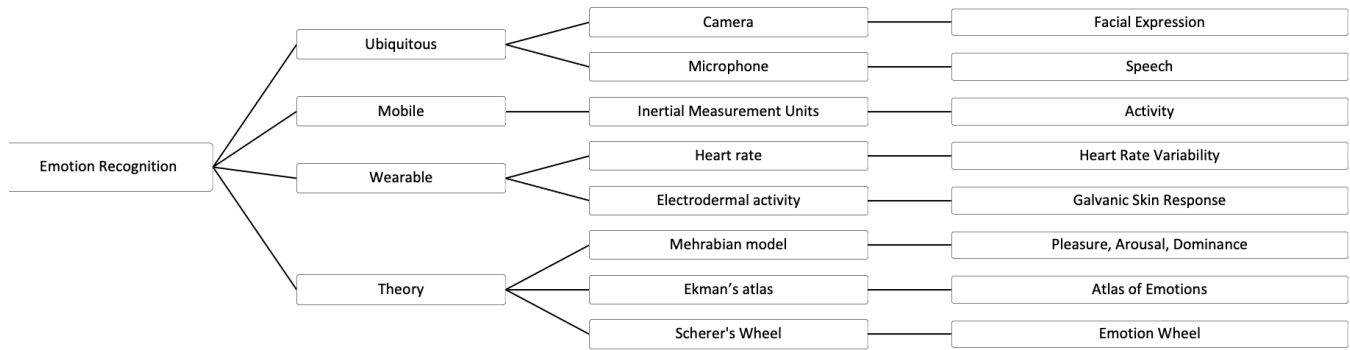


Figure 2: Design space of emotion recognition: an overview of platforms, theoretical models, sensors, and data

2.3 Approaches: Sensors and Data

Emotion recognition systems focus on traditional affect detection modalities like physiology, face, and voice, or analyze data sources such as text (e.g. through social media posts), body language, and complex multimodal systems [6].

The physiological signals traditionally analyzed include heart rate (ECG, PPG, BVP, HRV), galvanic skin response (electrodermal activity, skin conductance), EEG, fNIRS, and speech production. Among the activity sensors, inertial measurement units such as: accelerometers, gyroscope, magnetometers or compass have been largely explored, mostly to detect users' gestures or behaviors and activities. Facial expressions are analyzed in a contact-less approach through video cameras and image processing, but can also benefit from electrodes placed on users' skin, in a more obtrusive approach. Approaches that rely on eye tracking, for analysis of gaze and dwell time, require initial calibration to enhance accuracy of the detection results.

With microphones widely available in mobile devices, including wearables, smartphones, and intelligent personal assistants, the attempts to analyze speech and detect users' emotions have become widespread. To that, systems rely on artificial intelligence and machine learning models to analyze and extract several features of speech and acoustics, including pitch, jitter, energy, rate, and length and number of pauses [28]. Changes in speech are one of the physiological changes that have been proven effective to indicate stress [14].

In addition to physiological sensors, environmental sensors, including those that collect data about users' location, time, weather, and temperature have been also been used for emotion recognition. They are coupled with information from users' data in hybrid solutions. The combination of multiple devices, sensors, and data signals aim to improve the accuracy of emotion recognition systems by taking a more holistic approach. In practice such systems are more costly to implement, however with advances in technology, large scale production and adoption, and optimized models, it is expected that future methods will become optimized and as such benefit from combined approaches [28].

This paper does not focus on social media analytics, including mainly pervasive solutions. However, it is worthy noting that a

growing body of research has investigated how users express emotions using natural language processing, and how sentiment analysis, and text analytics contribute to detect emotions [2, 7, 8]. Such approaches are inexpensive to implement, however they also pose additional risks for users' privacy, besides raising ethical concerns [30].

2.4 Applications

A wide range of applications can benefit from emotion recognition—especially solutions that are unobtrusive, accurate, and responsive. In art installations, the user experience can be tailored to engage and intrigue users. In Education, the mental health and students' behaviors have been assessed using smart phones and self-reports [12, 29]. Such understanding is important to identify the drivers to depression, helping higher education programs to better serve college students, by detecting depressive symptoms early and preventing depression [31]. In the automotive industry, several studies sought to detect whether the drivers feel stressed, anxious, or angry. Such studies instrument the steering wheel of the car or the driver's seat with sensors [23, 32]. Results show that stress can be detected quickly when analyzing driving patterns [24].

3 DISCUSSION

Even though emotion recognition using alternative approaches is now feasible and relatively inexpensive, accuracy and reliability are well-known limitations. The choice of approach must take into account their specific drawbacks and benefits. Moreover, knowing domain-specific and user requirements is crucial to ensure that recognition models are enough accurate in the classification process. Potential risks for users' privacy and safety must be reduced, which requires in-depth discussions and reflections about the opportunities and challenges involved, as well as the benefits and risks that are specific to each application. More broadly, open questions remain about how to address such challenges. Hence, a critical debate is needed to discuss how to implement and deploy systems that are optimized to collect only essential data and as such preserve users' privacy. Open questions remain regarding how to provide accurate classifications, to minimize risks, and to personalize interventions to meet individual's needs.

Design considerations must involve questions about fairness, ethics, and the validity and reliability of data and classification

models. There are many serious questions about ethics and how data can be used for users' benefits and to minimize risks. Regarding privacy, the fact that people wear and carry devices that continuously sense data about them, raises privacy concerns. The decisions about data collection, storage, usage, and disclosure must be informed. Users must not only be aware about what is collected, but also how the information is used, to prevent that sensitive data is disclosed to unauthorized parties [13]. Regarding instrumented environments and smart devices, in the US, there is a risk that the device services disclose HIPAA-protected information to a nonauthorized person. To prevent that manufacturers need yet to spend time and efforts to ensure to users that their privacy is not at risk [28].

Emerging technologies are able to improve the accuracy of the data collection and information extracted or inferred at scale. The impact of predicting, recognizing, and intervening with emotions may reduce human risks, at the same time that, without proper regulation, it can be used to serve for accidental or nefarious harm [28]. Having a device actively listening to end users, implies that users' privacy can be compromised, meaning that stress levels as well as current health status could be revealed without an informed consent from the individual affected [28]. Lastly, it is unclear to what extent it is ethical to provide stressed individuals with information regarding nearby psychotherapy or antidepressant medication options without their explicit requests [28].

4 CONCLUSION

Emotion recognition has been increasingly explored for applications that monitor users, provide them with interventions, or adapt the system design to meet users' needs. Despite significant growth in the research and development of emotion recognition systems, current outcomes are limited in terms of accuracy, efficiency, applicability, among others. Several approaches have been investigated to recognize human emotions. More specifically, state-of-the-art technologies explore the analysis of facial expressions, speech, movement, and physiological signs. They combine data from mobile as well as environmental sensors. Each approach is customized to meet end users' and domain-specific requirements. To characterize the existing approach and shed light into their advantages and limitations, this paper presents a design space of emotion recognition, discussing the feasibility of each approach, as well as their characteristics. We conclude with a critical analysis of implications for privacy and fairness.

ACKNOWLEDGMENTS

This research received financial support from the National Institute on Disability, Independent Living, and Rehabilitation Research (NIDILRR) under Grant No. 90DPGE0009.

REFERENCES

- [1] Nazanin Andalibi. 2020. Disclosure, privacy, and stigma on social media: Examining non-disclosure of distressing experiences. *ACM transactions on computer-human interaction (TOCHI)* 27, 3 (2020), 1–43.
- [2] Nazanin Andalibi and Justin Buss. 2020. The human in emotion recognition on social media: Attitudes, outcomes, risks. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [3] Nazanin Andalibi and Madison K Flood. 2021. Considerations in Designing Digital Peer Support for Mental Health: Interview Study Among Users of a Digital Support System (Buddy Project). *JMIR mental health* 8, 1 (2021), e21819.
- [4] Jakob E Bardram, Mads Frost, Károly Szántó, and Gabriela Marcu. 2012. The MONARCA self-assessment system: a persuasive personal monitoring system for bipolar patients. In *Proceedings of the 2nd ACM SIGHIT International Health Informatics Symposium*. 21–30.
- [5] Lisa Feldman Barrett. 2017. *How emotions are made: The secret life of the brain*. Houghton Mifflin Harcourt.
- [6] Rafael A Calvo and Sidney D'Mello. 2010. Affect detection: An interdisciplinary review of models, methods, and their applications. *IEEE Transactions on affective computing* 1, 1 (2010), 18–37.
- [7] Munmun De Choudhury, Scott Counts, and Michael Gamon. 2012. Not all moods are created equal! exploring human emotional states in social media. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 6.
- [8] Munmun De Choudhury, Michael Gamon, Scott Counts, and Eric Horvitz. 2013. Predicting depression via social media. In *Seventh international AAAI conference on weblogs and social media*.
- [9] Paul Ekman. 2018. *Atlas of Emotion*.
- [10] Charles Fage. 2015. An emotion regulation app for school inclusion of children with ASD: design principles and preliminary results for its evaluation. *ACM SIGACCESS Accessibility and Computing* 112 (2015), 8–15.
- [11] Stephen W Gilroy, Marc Cavazza, Rémi Chaignon, Satu-Marja Mäkelä, Markus Niranen, Elisabeth André, Thirid Vogt, Jérôme Urbain, Hartmut Seichter, Mark Billingham, et al. 2008. An affective model of user experience for interactive art. In *Proceedings of the 2008 international conference on advances in computer entertainment technology*. 107–110.
- [12] Jeremy F Huckins, Alex W DaSilva, Weichen Wang, Elin Hedlund, Courtney Rogers, Subigya K Nepal, Jialing Wu, Mikio Obuchi, Ellis I Murphy, Meghan L Meyer, et al. 2020. Mental health and behavior of college students during the early phases of the COVID-19 pandemic: Longitudinal smartphone and ecological momentary assessment study. *Journal of medical Internet research* 22, 6 (2020), e20185.
- [13] Predrag Klasnja, Sunny Consolvo, Tanzeem Choudhury, Richard Beckwith, and Jeffrey Hightower. 2009. Exploring privacy concerns about personal sensing. In *International Conference on Pervasive Computing*. Springer, 176–183.
- [14] Hong Lu, Denise Fraundorfer, Mashfiqui Rabbi, Marianne Schmid Mast, Gokul T Chittaranjan, Andrew T Campbell, Daniel Gatica-Perez, and Tanzeem Choudhury. 2012. Stresssense: Detecting stress in unconstrained acoustic environments using smartphones. In *Proceedings of the 2012 ACM conference on ubiquitous computing*. 351–360.
- [15] Regan L Mandryk, Kori M Inkpen, and Thomas W Calvert. 2006. Using psychophysiological techniques to measure user experience with entertainment technologies. *Behaviour & information technology* 25, 2 (2006), 141–158.
- [16] Conor McGinn and Kevin Kelly. 2018. Using the Geneva Emotion Wheel to Classify the Expression of Emotion on Robots. In *Companion of the 2018 ACM/IEEE International Conference on Human-Robot Interaction*. 191–192.
- [17] Albert Mehrabian. 1995. Framework for a comprehensive description and measurement of emotional states. *Genetic, social, and general psychology monographs* (1995).
- [18] Varun Mishra. 2019. From sensing to intervention for mental and behavioral health. In *Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers*. 388–392.
- [19] Varun Mishra, Sougata Sen, Grace Chen, Tian Hao, Jeffrey Rogers, Ching-Hua Chen, and David Kotz. 2020. Evaluating the Reproducibility of Physiological Stress Detection Models. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 4 (2020), 1–29.
- [20] Vivian Genaro Motti. 2020. *Design Considerations*. Springer International Publishing, Cham, 41–79. https://doi.org/10.1007/978-3-030-27111-4_2
- [21] Vivian Genaro Motti. 2020. Introduction to wearable computers. In *Wearable Interaction*. Springer, 1–39.
- [22] Vivian Genaro Motti and Kelly Caine. 2014. Human factors considerations in the design of wearable devices. In *Proceedings of the human factors and ergonomics society annual meeting*, Vol. 58. SAGE Publications Sage CA: Los Angeles, CA, 1820–1824.
- [23] Pablo Paredes, David Sun, and John Canny. 2013. Sensor-less sensing for affective computing and stress management technology. In *2013 7th International Conference on Pervasive Computing Technologies for Healthcare and Workshops*. IEEE, 459–463.
- [24] Pablo E Paredes, Francisco Ordonez, Wendy Ju, and James A Landay. 2018. Fast & furious: detecting stress with a car steering wheel. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [25] Pablo E Paredes, Yijun Zhou, Nur Al-Huda Hamdan, Stephanie Balters, Elizabeth Murnane, Wendy Ju, and James A Landay. 2018. Just breathe: In-car interventions for guided slow breathing. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 1 (2018), 1–23.
- [26] Mashfiqui Rabbi, Shahid Ali, Tanzeem Choudhury, and Ethan Berke. 2011. Passive and in-situ assessment of mental and physical well-being using mobile sensors. In *Proceedings of the 13th international conference on Ubiquitous computing*. 385–394.

- [27] Klaus R Scherer. 2005. What are emotions? And how can they be measured? *Social science information* 44, 4 (2005), 695–729.
- [28] George M Slavich, Sara Taylor, and Rosalind W Picard. 2019. Stress measurement using speech: Recent advancements, validation issues, and ethical and privacy considerations. *Stress* 22, 4 (2019), 408–413.
- [29] Rui Wang, Fanglin Chen, Zhenyu Chen, Tianxing Li, Gabriella Harari, Stefanie Tignor, Xia Zhou, Dror Ben-Zeev, and Andrew T Campbell. 2014. StudentLife: assessing mental health, academic performance and behavioral trends of college students using smartphones. In *Proceedings of the 2014 ACM international joint conference on pervasive and ubiquitous computing*. 3–14.
- [30] Ken Ward. 2018. Social networks, the 2016 US presidential election, and Kantian ethics: applying the categorical imperative to Cambridge Analytica’s behavioral microtargeting. *Journal of media ethics* 33, 3 (2018), 133–148.
- [31] Xuhai Xu, Prerna Chikersal, Afsaneh Doryab, Daniella K Villalba, Janine M Dutcher, Michael J Tumminia, Tim Althoff, Sheldon Cohen, Kasey G Creswell, J David Creswell, et al. 2019. Leveraging routine behavior and contextually-filtered features for depression detection among college students. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 3 (2019), 1–33.
- [32] Sebastian Zepf, Monique Dittrich, Javier Hernandez, and Alexander Schmitt. 2019. Towards empathetic car interfaces: Emotional triggers while driving. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [33] Hui Zheng, Vivian Genaro Motti, Kudirat Giwa-Lawal, Anna Evmenova, and Heidi Graff. 2019. Evaluating WELL: A Wrist-Worn Application to Assist Young Adults with Neurodevelopmental Disorders in Inclusive Classes. In *IFIP Conference on Human-Computer Interaction*. Springer, 114–134.